

Learn to Denoise

DATA 443/643 · Week 12, note 12.2

Week 12

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Ideas from Stanford CS231N 2025 L14 (predict the noise; T reverse steps). Original slides; that course is not copied.

Lecture plan

- 1 MSE on ϵ , not an adversary
- 2 One reverse-step mean
- 3 Oracle: true ϵ peels back to x_0
- 4 T net calls versus one GAN pass

Key goal: the net predicts the noise you added in 12.1.

First time: what / why / formula

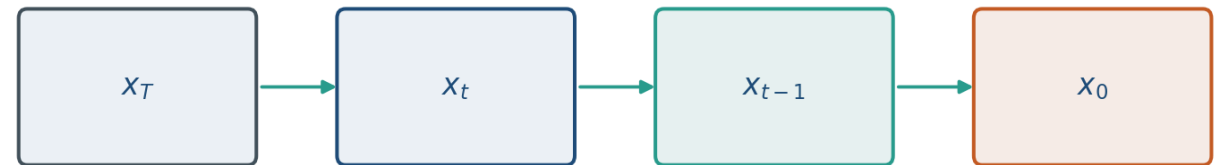
- What / why. A net $\varepsilon_\theta(x_t, t)$ predicts ε . Training is MSE on that noise. Sampling starts from $x_T \sim \mathcal{N}(0, I)$ and denoises. You cannot write the true reverse in closed form for images. Predicting ε is equivalent (under the usual parameterization) to predicting the posterior mean.
- Architecture. U-Net or DiT on x_t with a time embedding. Lab 12 uses an oracle mean so you see the geometry without training.
- Formula. x_{t-1} from ε_θ via the standard DDPM mean; cartoon: subtract a scaled ε_θ and add a little noise (except $t = 1$).
- Tradeoffs. + MSE is easier than GAN min-max. – Many steps at sample time; quality vs speed; classifier-free guidance is next note.

How. Loss: $\mathbb{E} \|\varepsilon - \varepsilon_\theta(x_t, t)\|^2$. Sample: T reverse steps (or fewer with a faster sampler, later papers).

Ordinary regression

- Sample t , form x_t , guess ε .
- Predicting ε is equivalent (up to scale) to x_0 or the score.
- t is an input. The same x_t is a different problem at $t=10$ and $t=900$.

Reverse: a network predicts the noise (or x_0) and steps toward data



DDPM training loss

$$\mathcal{L} = \|\varepsilon - \varepsilon_{\theta}(x_t, t)\|^2$$

No discriminator. A known target.

Sampling still walks $t = T \dots 1$.

Lab 12 uses the true posterior mean.
You know x_0 .

Reuse $x_t = 2.0$ from 12.1

Oracle $\varepsilon_\theta = \varepsilon = 2$: one reverse mean lands on $x_0 = 1$.

- $\alpha_t = \bar{\alpha}_t = 0.64$, $\varepsilon_\theta = 2$.
- The reverse mean lands on $x_0 = 1.0$.
- If $\varepsilon_\theta = 0$, you mostly rescale x_t . Not a sample.

piece	value
x_t	2.0
$(1 - \alpha_t)/\sqrt{1 - \bar{\alpha}_t} \cdot \varepsilon$	1.2
x_{t-1} mean	1.0

The trade

GAN (Week 11)

One forward pass. Min-max. Collapse is the usual failure.

DDPM

T network calls (or fewer with DDIM). Report the sampler next to the metric.

What to take from this lecture

Oracle

Lab 12 does not train a U-Net. It uses the known x_0 .

Noise on the way down

Drop $\sigma_t z$ and every sample looks the same.

Report T

Changing T at test time is part of the method.